

Chapter 2

QBIO 305 FA26

Summary of lecture slides

Chapter 2

- Variables
 - Quantitative: discrete, continuous
 - Qualitative: categorical, ordinal
- Median
- Mean
- IQR
- 5 number summary
 - Minimum, Q1, median, Q3, maximum
- Robustness

Variable

- Characteristic of a person or thing that can be assigned a number or category
- Categorical variable
 - No obvious order
 - E.g. blood type, gender
- Numeric variable
 - Discrete: integers, no decimals/fractions
 - E.g. Number of children a mother has, length of DNA in basepairs
 - Continuous: decimals/fractions allowed
 - E.g. weight of a baby, length of a snake

Observational unit

- Sometimes n persons are sampled and multiple variables are collected for each
 - E.g. blood type, gender, and weight of 300 babies are collected
 - The baby is called the “observational unit”

- Frequency distribution
- Figures
 - Always label axes large enough to read
 - When comparing multiple figures, check the axes
 - Units may be different, numbers may be drastically different
- Relative frequency = $\text{count} / \text{sample size}$

Bar charts/histograms

- Area of one or several bars is proportional to the corresponding frequency

Grouped frequency distribution

- Continuous variables require a decision of how to group the data
- E.g. grouping weight for children:
 - The distribution for 5 lb bins may look different than 10 lb bins

Distribution shapes

- Bell-shaped (or Gaussian or normal)
- Bimodal (2 peaks)
- Asymmetric
 - Skewed to the right (outlier is on right side)
 - Skewed to the left
- Exponential

Descriptive statistics

- A statistic is a numerical measure calculated from sample data
- Median
 - The value that most nearly lies in the middle of the sample
 - If the sample size is even, we average the 2 middle numbers
- Mean
 - The sum of the samples divided by the sample size

Robustness

- A statistic is robust if it is relatively unaffected by changes in a small portion of the data
- The median is robust and the mean is not
- Trimmed mean:
 - Removing a percentage of the highest and lowest data points before calculating the mean

Boxplots

- Median splits distribution into two parts (lower and upper)
- The quartiles split each of the parts in half
 - The first quartile (Q1) splits the lower half
 - The third quartile (Q3) splits the upper half
- Interquartile range (IQR)
 - The difference between the third and first quartiles
 - This measures the spread of the middle 50% of the distribution

Boxplots

- Boxplot with no outliers:
 - 5 number summary: minimum, Q1, median, Q3, maximum
- Box plot with outliers:
 - Lower fence = $Q1 - (1.5 \times IQR)$
 - Upper fence = $Q3 + (1.5 \times IQR)$
 - Any data point lower than the lower fence or higher than the upper fence is an outlier
 - Whiskers extend to smallest and largest data points that are not outliers
 - Outliers are identified by dots

Multivariate data

- Consider the relationship between multiple variables

Big data

- Overwhelming amount of data
- Difficult to summarize and visualize

Standard deviation

$$s = \sqrt{\frac{\sum_i (y_i - \bar{y})^2}{n - 1}}$$

Where $\bar{y}$ is the mean

Linear transformations $a+bX$

- Median/Mean
 - If we add a number, add it to the median
 - If we multiply by a number, multiply the median by that number
 - Also applies to Q1 and Q3
 - IQR/sd: only the multiplication affects this statistic, the addition cancels

Coefficient of variation

- sd/mean
- Same ratio for all scales